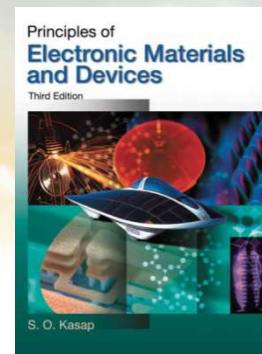




南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



Ch1-5 Elementary Materials Science Concepts

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Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

1.12.1 Isomorphous Solid Solutions 同构固溶体

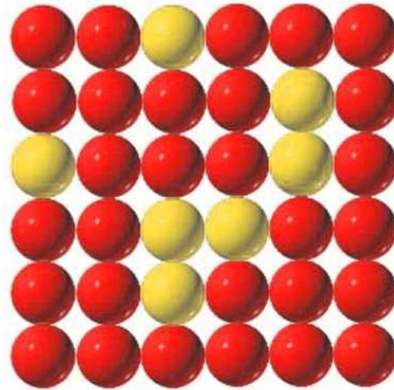
A **phase** of a material has the same composition, structure, and property everywhere.

Many solids are a homogeneous mixture of two types of separate atoms. For example, Ni-Cu alloy, is called a **solid solution**. The atoms in the majority make up the **solvent**, whereas the atoms in the minority are the **solute**, which is dissolved in the solvent.

Classification 1) According to the position of the solute atom in the lattice

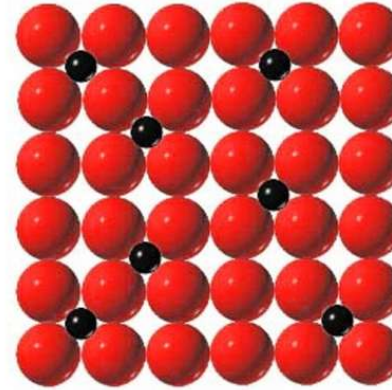
Substitutional
solid solution
取代固溶体

e.g. Cu-Ni



Interstitial
solid solution
填隙固溶体

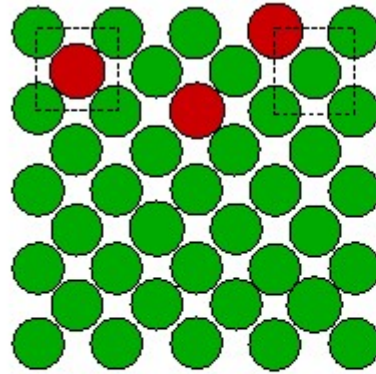
e. g. C-Fe



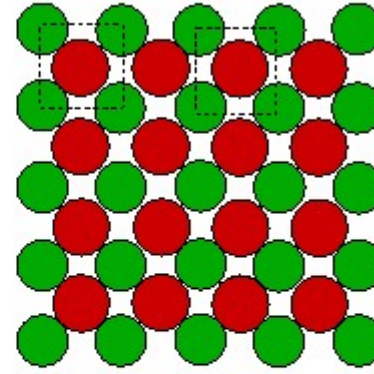
Most of the solids are substitutional solid solution



Substitutional solid solution 取代固溶体



(a) **Disordered** Substitutional Solid Solution. Example: Cu-Ni alloys ($\{100\}$ planes)



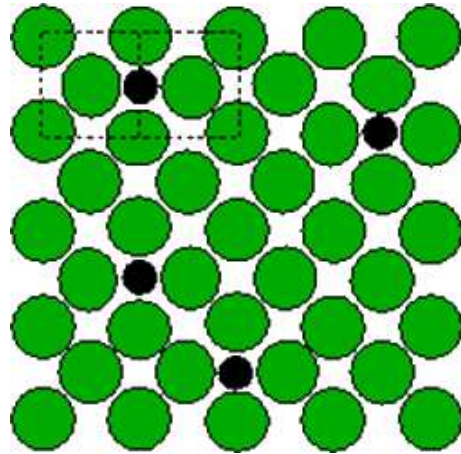
(b) **Ordered** Substitutional Solid Solution. Example: Cu-Zn alloy of composition 50%Cu-50%Zn. ($\{110\}$ planes).

Isomorphous solid solution 同构固溶体（置换固溶体）

- Metal oxide solid solution mainly occurs at the position of metal ion, e.g. MgO-CaO, MgO-CoO, PbZrO₃-PbTiO₃, Al₂O₃-Cr₂O₃ etc.
- MgO and CoO have the same NaCl-type structure, and the ionic radius of Mg²⁺ and Co²⁺ is similar. For MgO-ZnO solid solution, the Mg²⁺ in MgO can be replaced by Co²⁺ to form infinitely substitutional solid solution



Interstitial solid solution 填隙固溶体



(c) Interstitial Solid Solution.
Example: Small number of C
atoms in FCC Fe (austenite).
($\{100\}$ planes)

Solids solutions can be disordered substitutional, order substitutional and interstitial substitutional. There is only one phase within the alloy which has the same composition, structure and properties everywhere.

- The impurity atoms enter the gap position of the solvent lattice
杂质原子进入溶剂晶格的间隙位置
- For example, in the α Fe-C solid solution, the carbon atoms are located in the octahedral gap of the BCC lattice of the iron atoms.
如，在Fe-C系的 α 固溶体中，碳原子就位于铁原子的BCC点阵的八面体间隙中



Classification 2) According to the solubility of solute

1. Infinite solid solution (continuous solid solution)

- The solute and the solvent can be mutually dissolved without limitation of ratio
- Such as MgO-CoO, it can be expressed as $(\text{Mg}_x\text{Co}_{1-x})\text{O}$, where $x = 0\sim 1$, that is, Mg in MgO can be partially or completely replaced by Co

2. Limited solid solution (discontinuous solid solution)

- Impurity atoms have a certain solubility limit in the solid solution. A limited solid solution is generally formed if the two crystals differ in structure or have a large difference in the two radii.
- Such as MgO-CaO, Mg^{2+} radius 0.72 Å, Ca^{2+} radius 1.00 Å



Hume-Rothery Rule for Infinite Solid Solution

1) Crystal structure is the same (晶体结构相同)

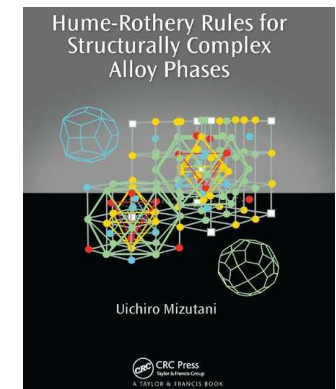
- The same crystal structure is necessary for the formation of an infinite solid solution. Different structures produce up to a limited solid solution

2) The difference between the two atoms is less than 15% (两种原子的大小相差小于15%)

- The dissolution of solute ions causes local distortion of the crystal lattice of the solvent. The solute ions are larger than the solvent ions, and the solute displaces the solvent ions around them. If the solute ions are smaller than the solvent ions, the surrounding solvent ions move closer to the solute ions.
- The larger the difference between the solute and the solvent size, the greater the degree of lattice distortion and the worse the crystal structure stability, which limits the further dissolution of the solute and reduces the solid solubility.

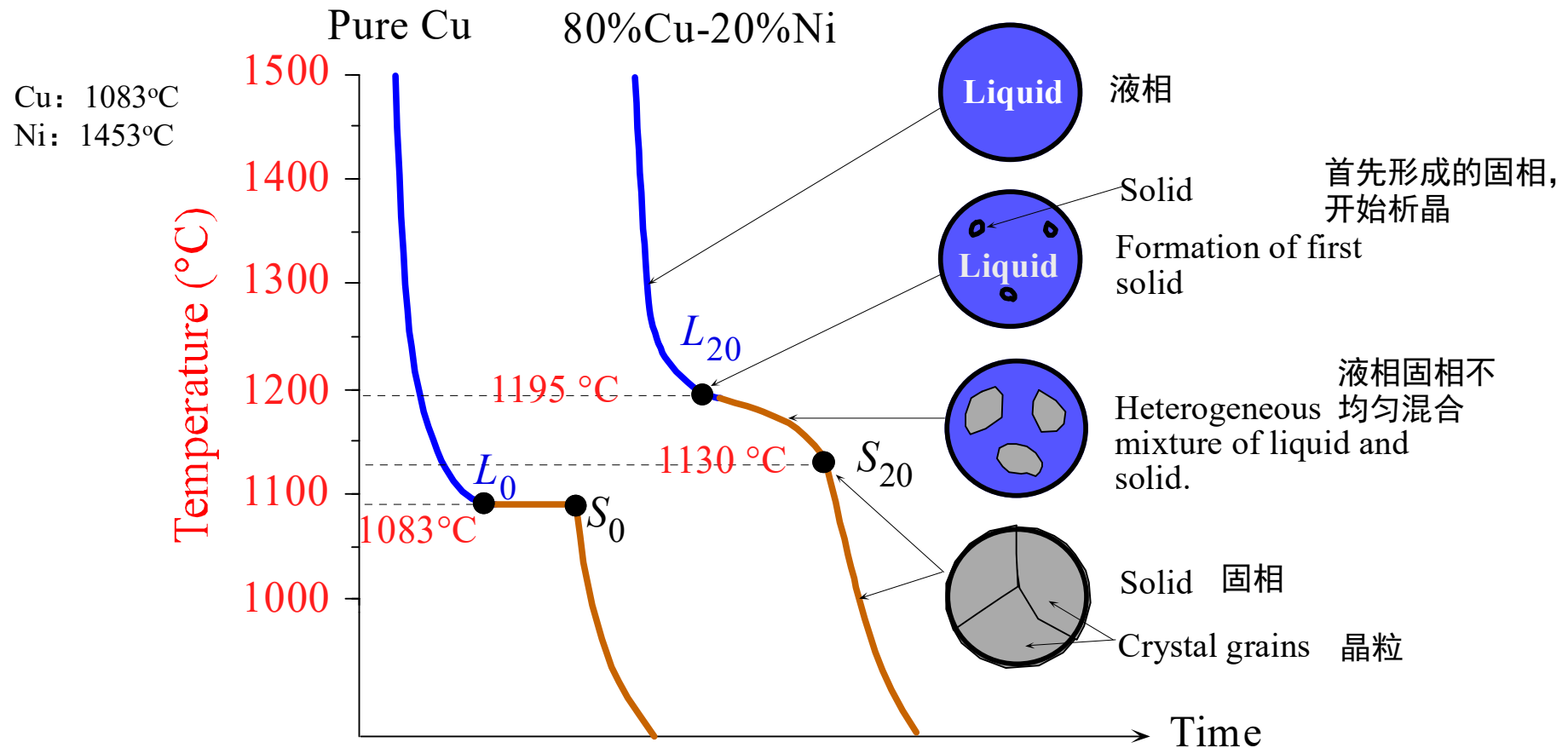
3) Atomic valence is the same (原子价相同)

4) Electronegativity is not much different (电负性相差不大)

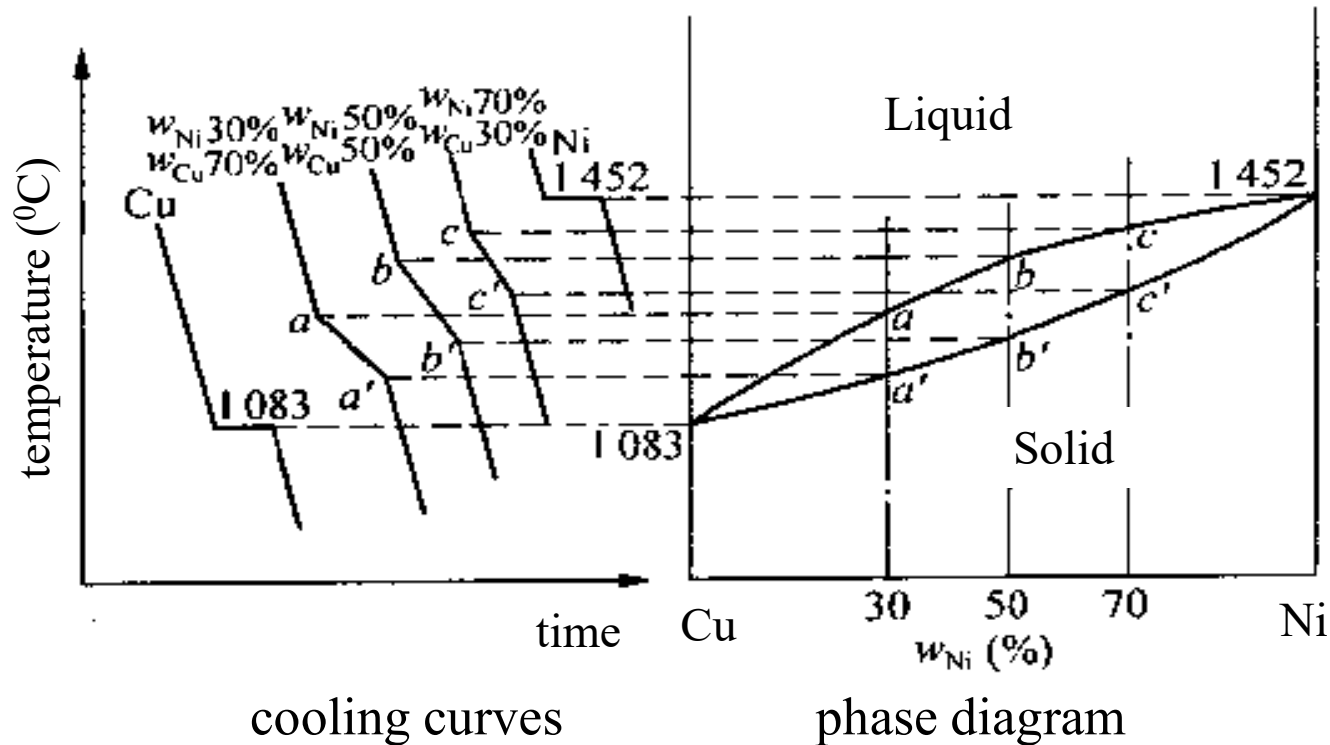


1.12.2 Phase diagrams: 相图

Solidification of an isomorphous alloy such as Cu-Ni. Typical cooling curves.



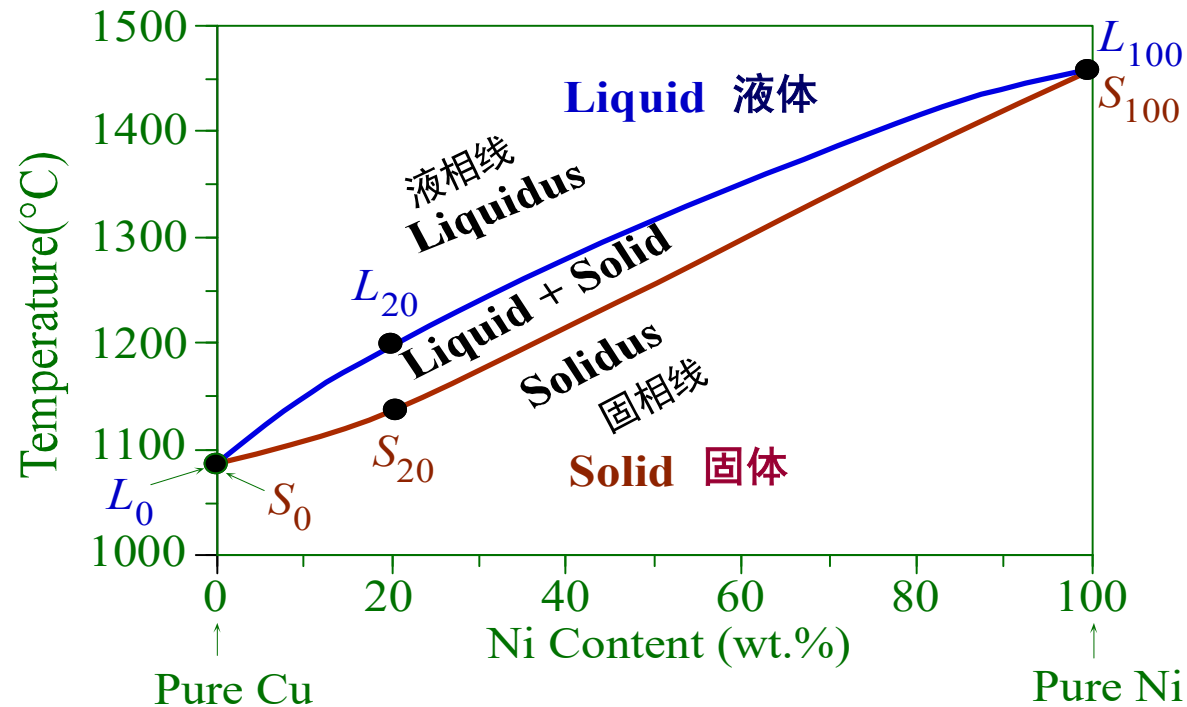
Phase diagrams: 相图



Application of phase diagrams: 1) Determine the composition of the material; 2) Develop material production & processing technology; 3) Analyze the organization of the equilibrium state and the changes of the imbalance state; 4) Predict material properties; 5) Failure analysis.



Phase diagrams: 相图

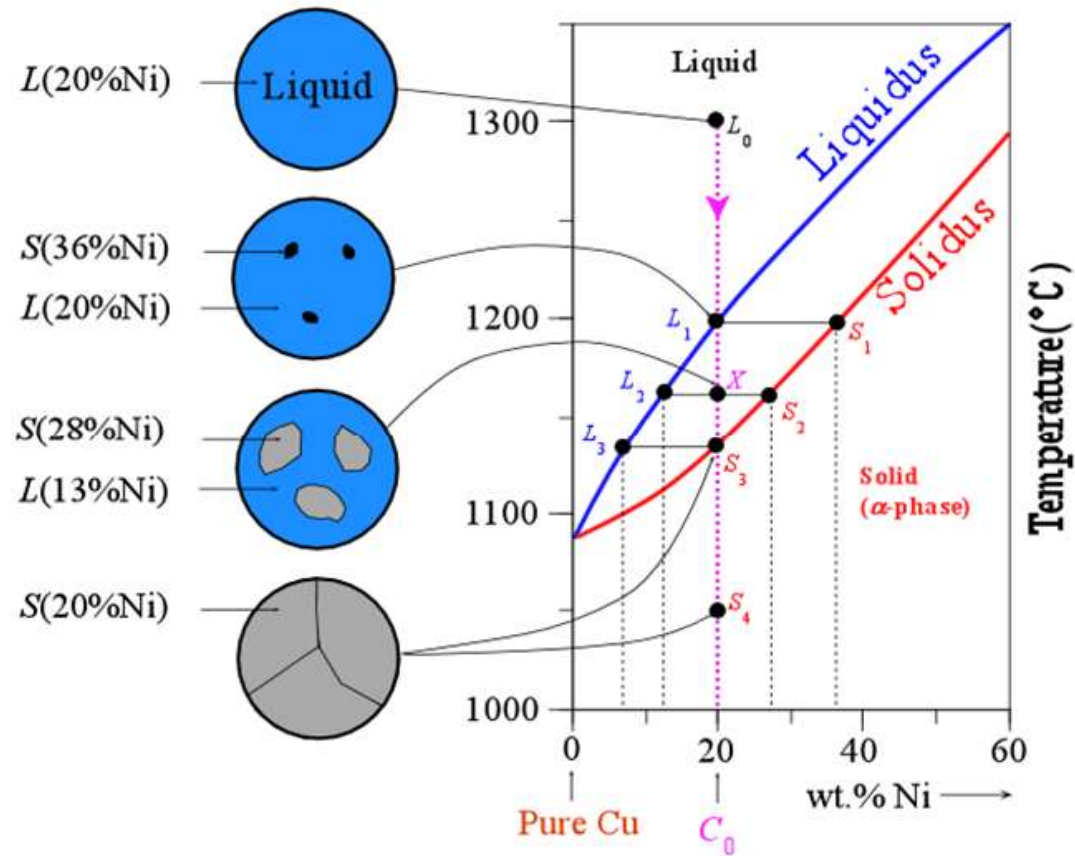


Liquidus curve : Link all the points corresponding to the beginning of freezing
液相线：表示合金结晶开始的温度或加热过程中熔化终了的温度

Solidus curve : Link all the points where the liquid has totally solidified
固相线：表示合金结晶终了的温度或在加热过程中开始熔化的温度



Phase diagrams: 相图



Cooling of a 80%Cu-20%Ni alloy from the melt to the solid state.



Phase diagrams: 相图

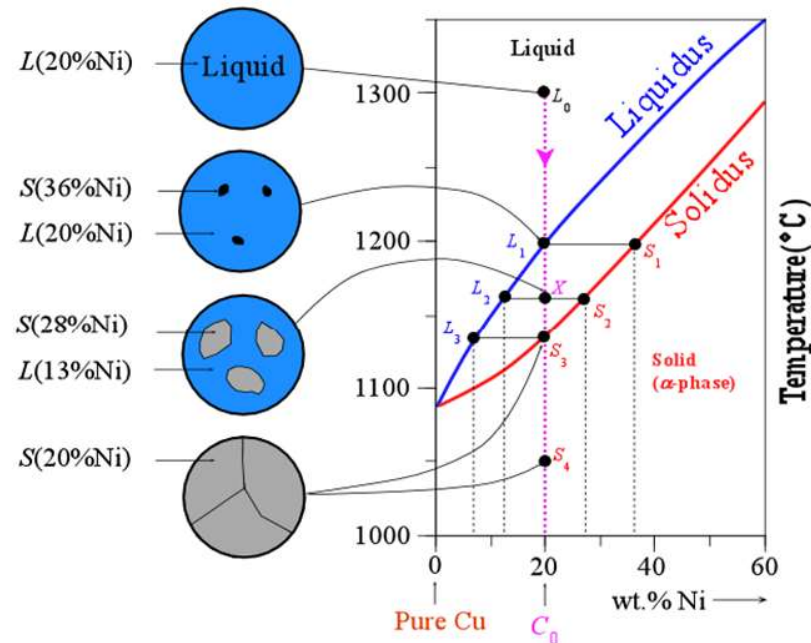


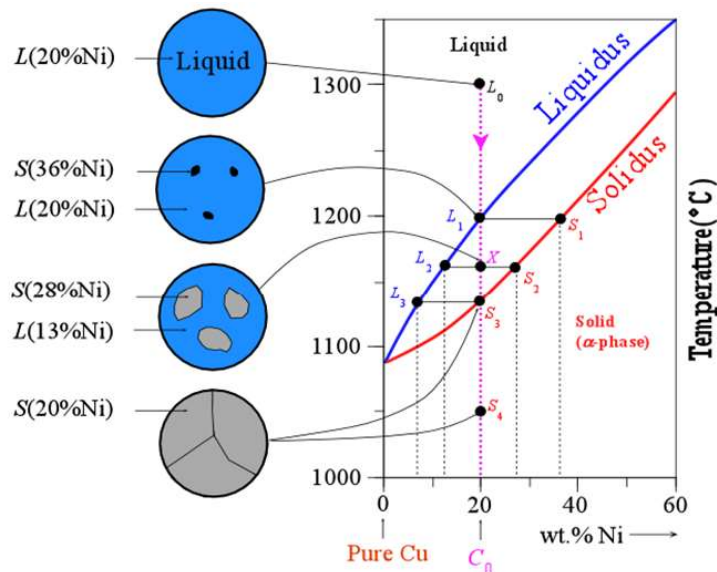
Table 1.6 Phase in the 80% Cu–20% Ni isomorphous alloy

Temperature, °C	Phases	Composition	Amount
1300	Liquid only	$L_0 = 20\% \text{ Ni}$	100%
1195	Liquid and solid	$L_1 = 20\% \text{ Ni}$ $S_1 = 36\% \text{ Ni}$	100% First solid appears
1160	Liquid and solid	$L_2 = 13\% \text{ Ni}$ $S_2 = 28\% \text{ Ni}$	w_L 53.3% w_α 46.7%
1130	Liquid and solid	$L_3 = 7\% \text{ Ni}$ $S_3 = 20\% \text{ Ni}$	The last liquid drop 100%
1050	Solid only	$S_4 = 20\% \text{ Ni}$	100%

weight fraction
(质量分数)



Level Rule (杠杆规则)



$$\begin{cases} W_L + W_S = 1 \\ C_L W_L + C_S W_S = C_O \end{cases}$$



$$W_L = \frac{C_S - C_O}{C_S - C_L}$$

and

$$W_S = \frac{C_O - C_L}{C_S - C_L}$$

[1.36]

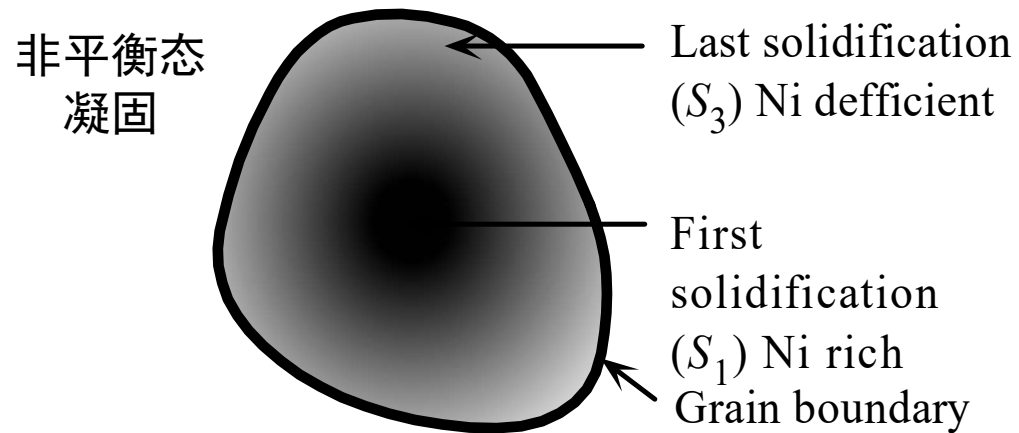
To apply Equation 1.36, we first draw a line (called a **tie line**) from L_2 to S_2 corresponding to C_L and C_S , as shown in Figure 1.64. The line represents a “horizontal lever” and point X at C_O at this temperature is the lever’s fulcrum. The lengths of the lever arms from the fulcrum to the liquidus and solidus curves are $(C_O - C_L)$ and $(C_S - C_O)$, respectively. The lever must be balanced by the weights W_L and W_S attached to the ends. The total length of the lever is $(C_S - C_L)$. At 1160 °C, $C_L = 0.13$ (13% Ni) and $C_S = 0.28$ (28% Ni), so the weight fraction of the liquid phase is

$$W_L = \frac{C_S - C_O}{C_S - C_L} = \frac{0.28 - 0.20}{0.28 - 0.13} = 0.533 \quad \text{or} \quad 53.3\%$$

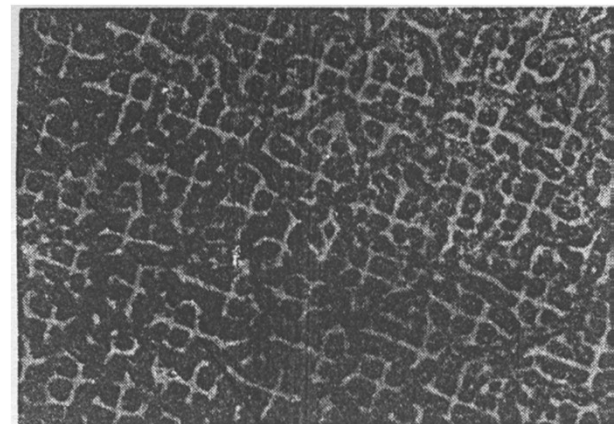
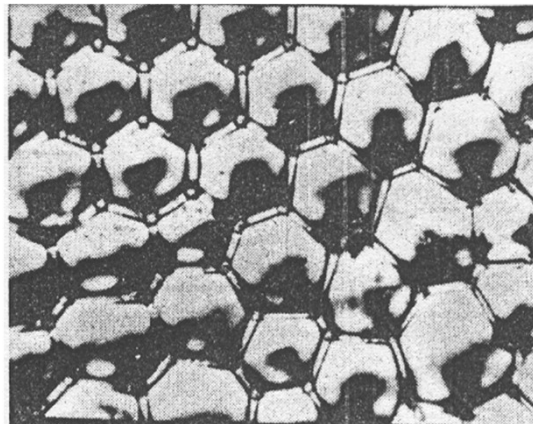
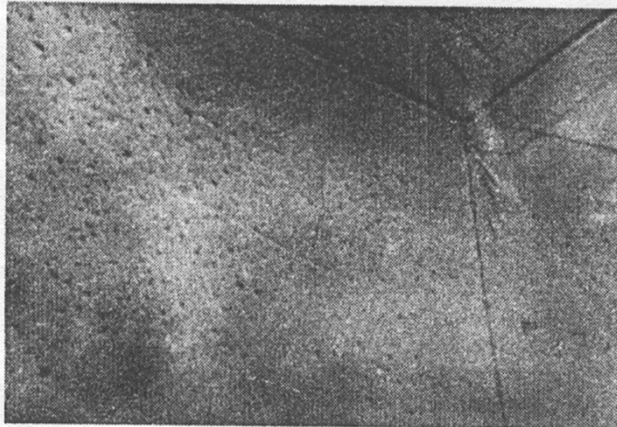
Similarly, the weight fraction of the solid phase is $1 - 0.533$ or 0.467.



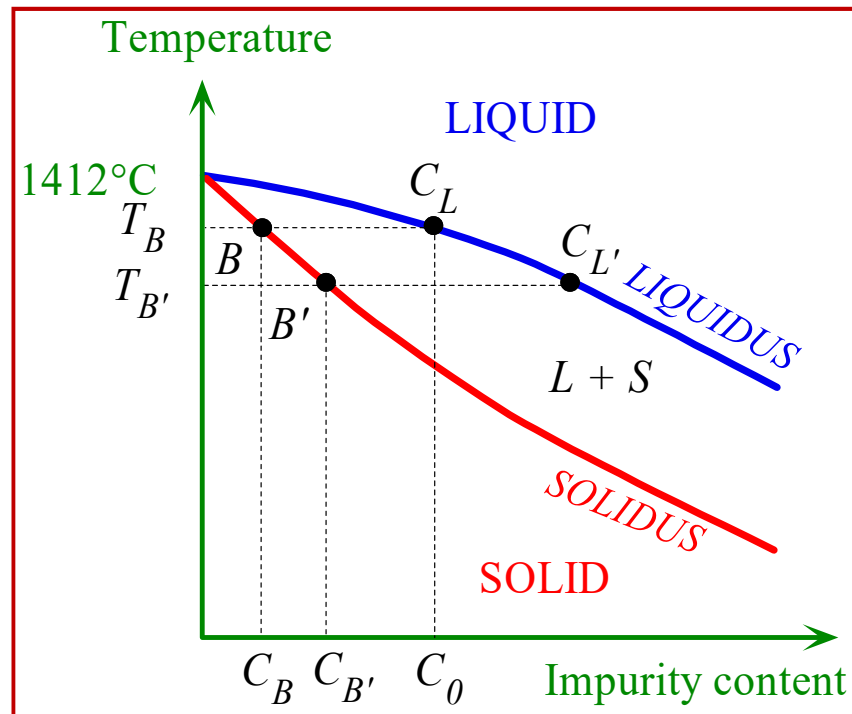
Segregation in a grain due to rapid cooling (non-equilibrium cooling)



Al-Cu alloy with different segregation geometries

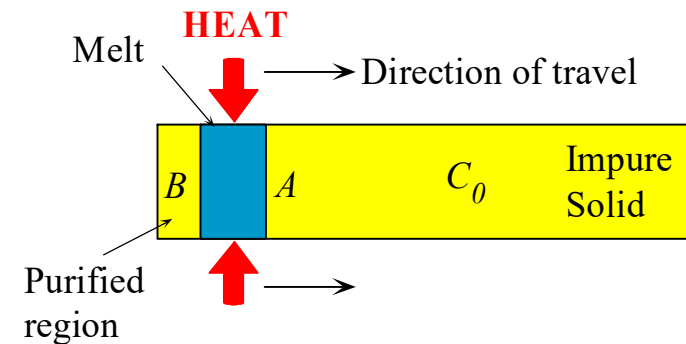


Zone refining and pure silicon crystals 区熔提纯和硅单晶的提纯



The phase diagram of Si with impurities near the low-concentration region

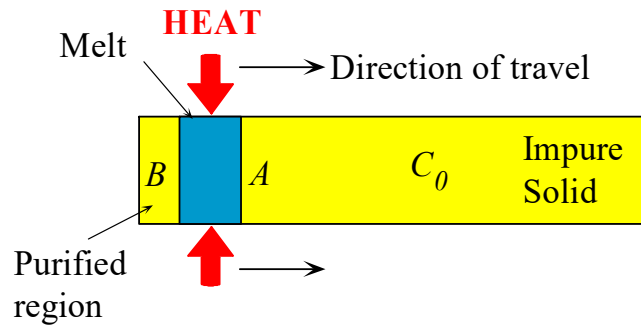
Silicon has a high melting temperature, so any impurities present in the crystal decrease the melting temperature



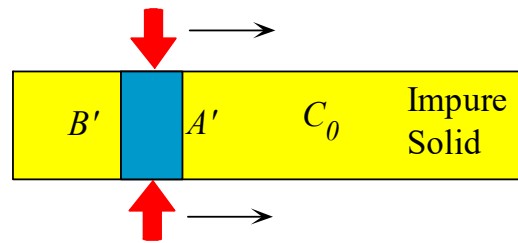
Zone refining is used for the production of high-purity crystals.



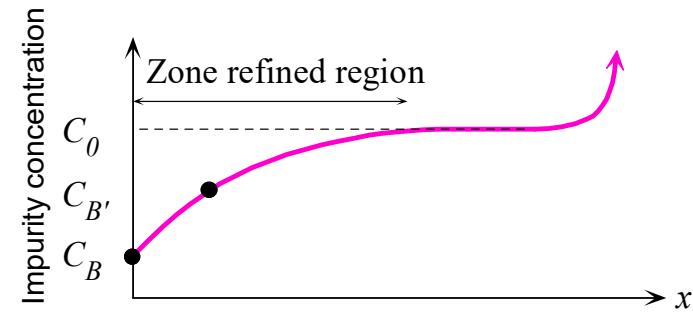
Zone refining 区熔提纯



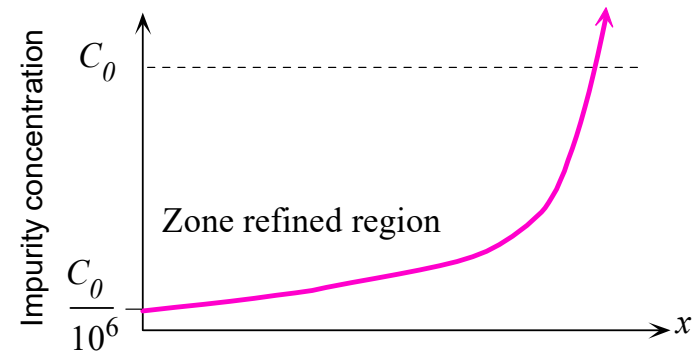
(a) Heat is applied locally starting at one end. The impurity concentration in the refrozen solid at B is $C_B < C_0$. The impurity concentration in the melt is $C_L' > C_0$.



(b) As the torch travels towards the right, the refrozen solid at B' has $C_{B'}$ where $C_B < C_{B'} < C_0$. The impurity concentration in the melt is now even greater than C_L' .



(c) The impurity concentration profile in the refrozen solid after one pass.

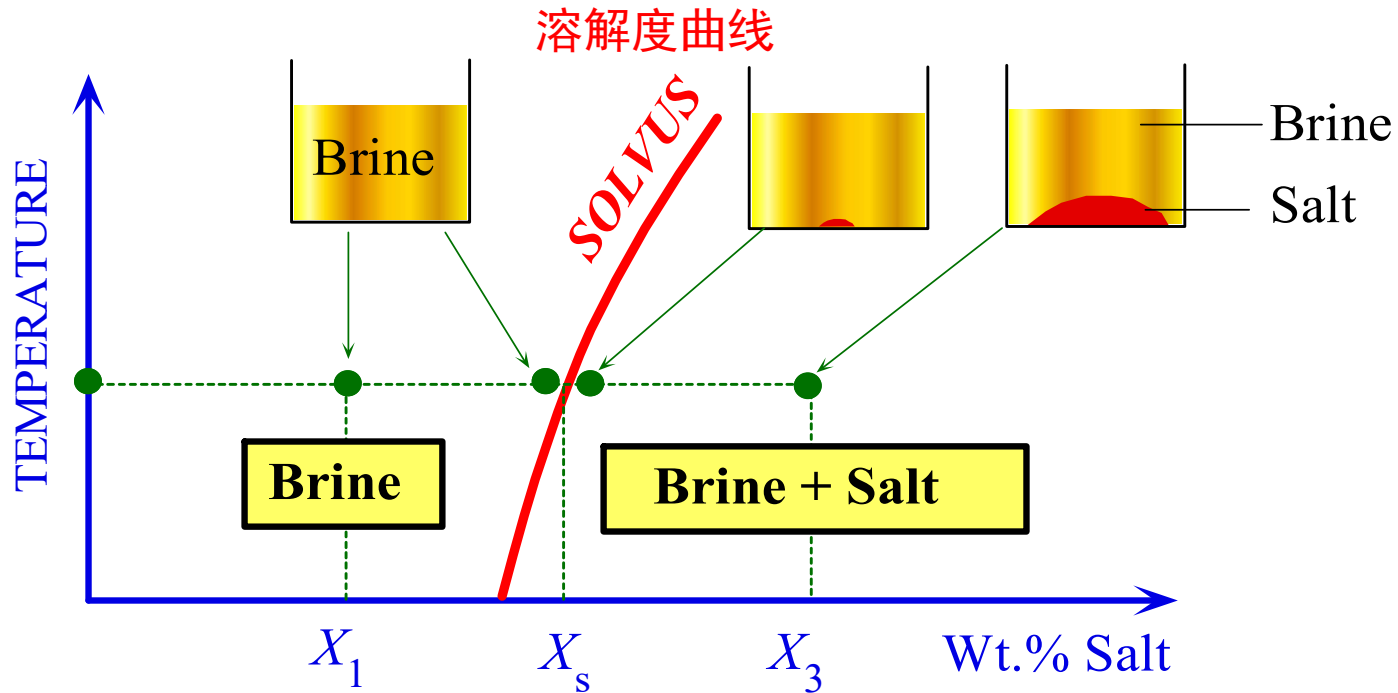


(d) Typical impurity concentration profile after many passes.



1.12.4 Binary Eutectic phase diagrams and Pb-Sn solders

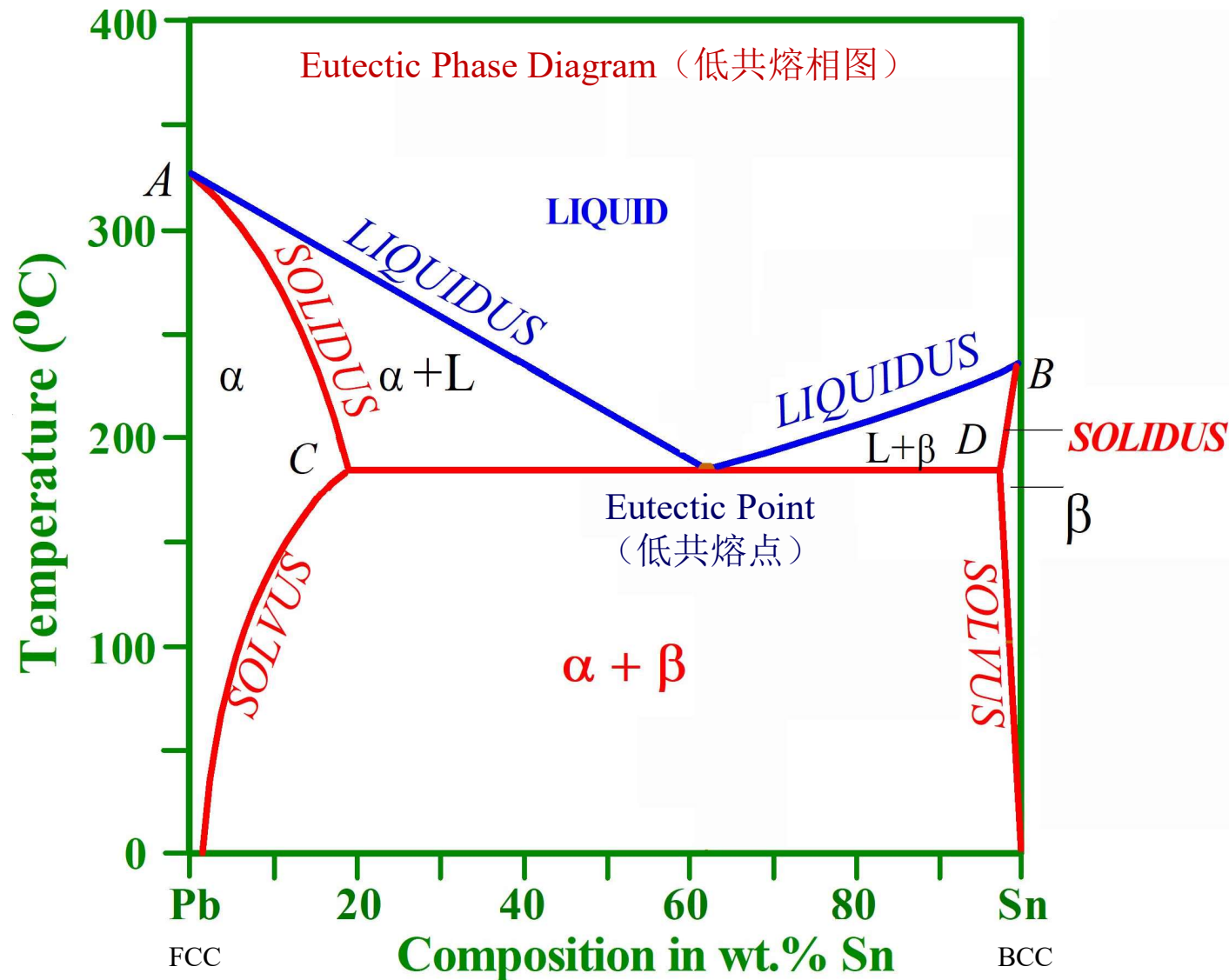
二元低共熔相图和Pb-Sn合金



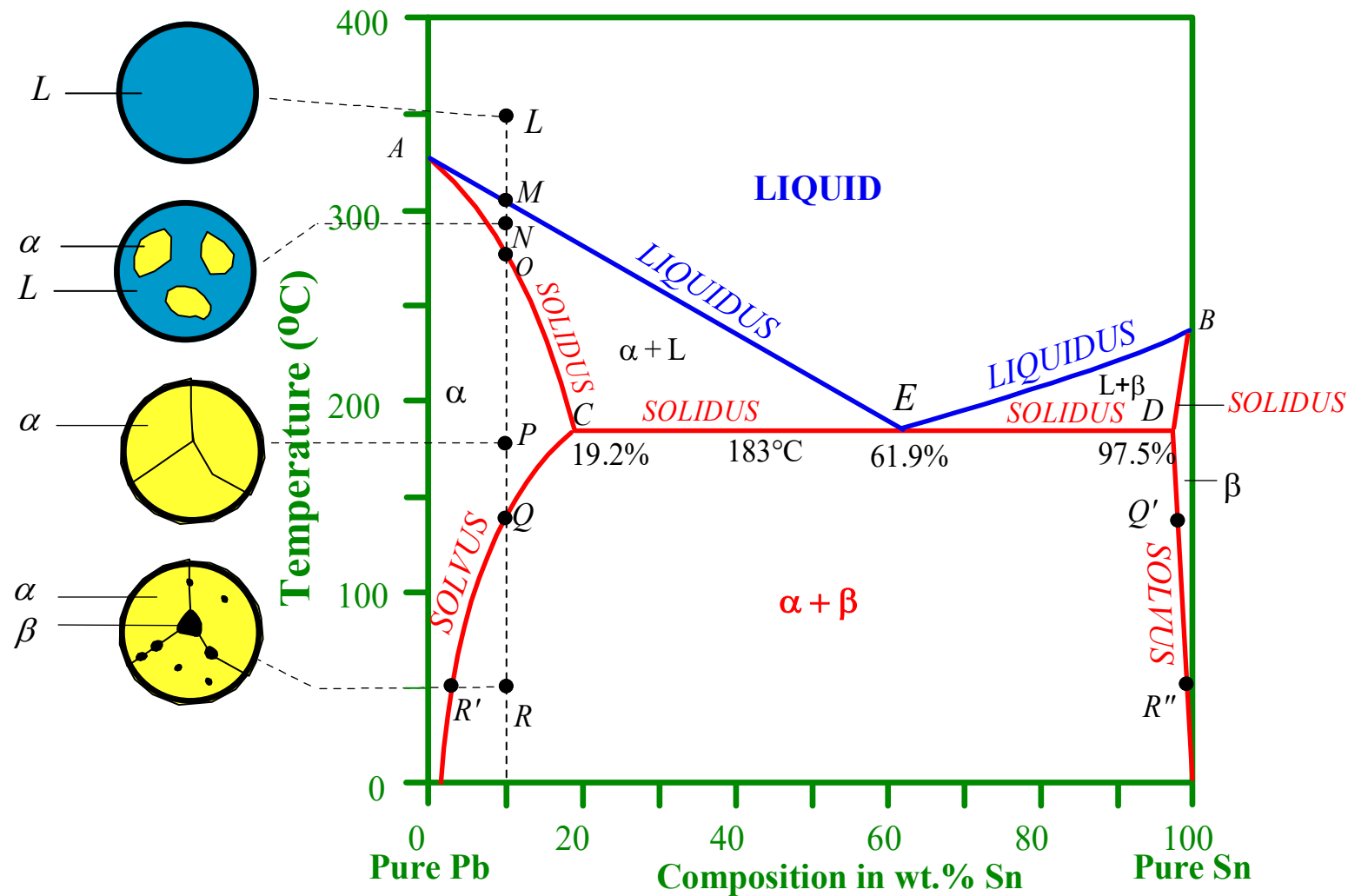
We can only dissolve so much salt in brine (solution of salt in water). Eventually we reach the solubility limit at X_s which depends on the temperature. If we add more salt then the excess salt does not dissolve and coexists with the brine. Past X_s we have two phases, brine (solution) and salt (solid).



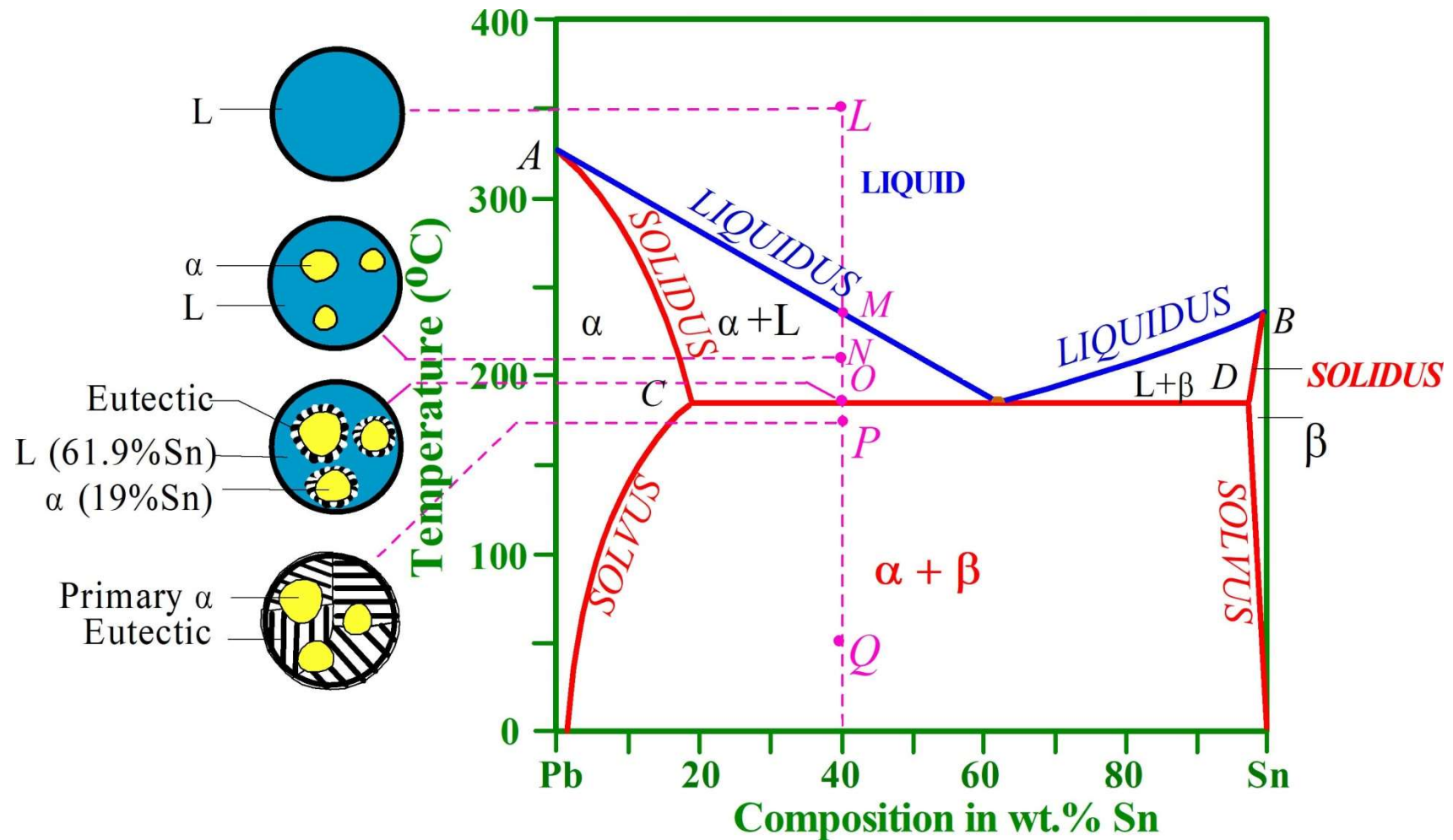
Equilibrium phase diagram of Pb-Sn alloy



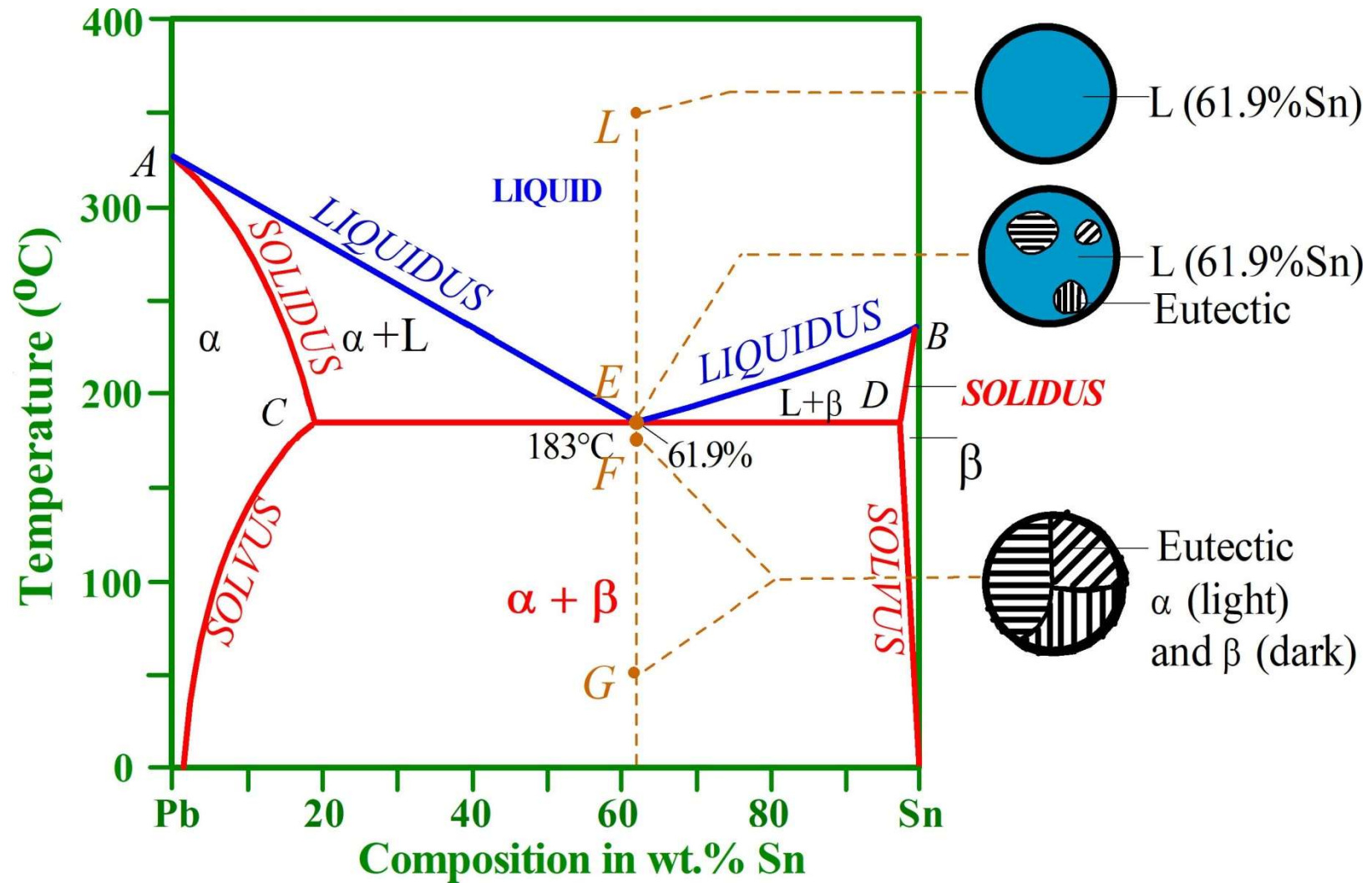
90% Pb-10% Sn Alloy



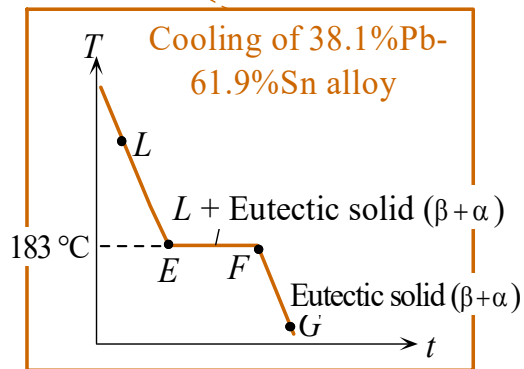
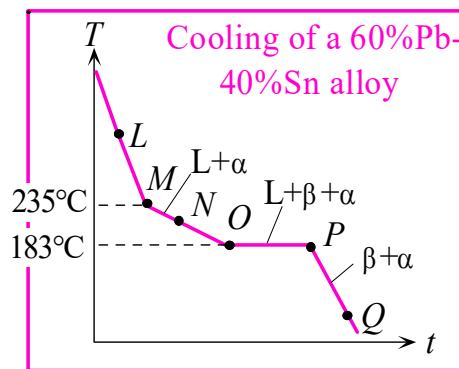
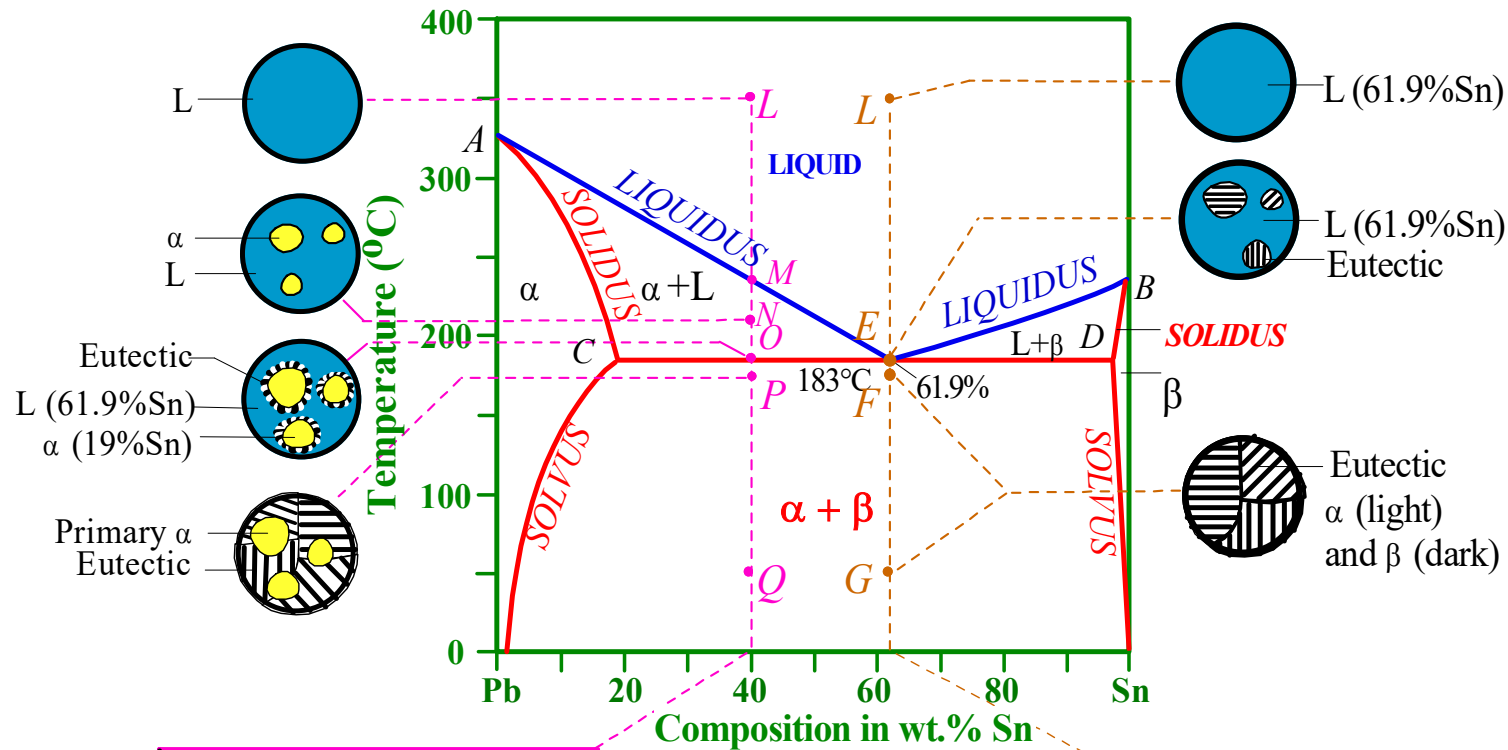
60% Pb-40% Sn Alloy

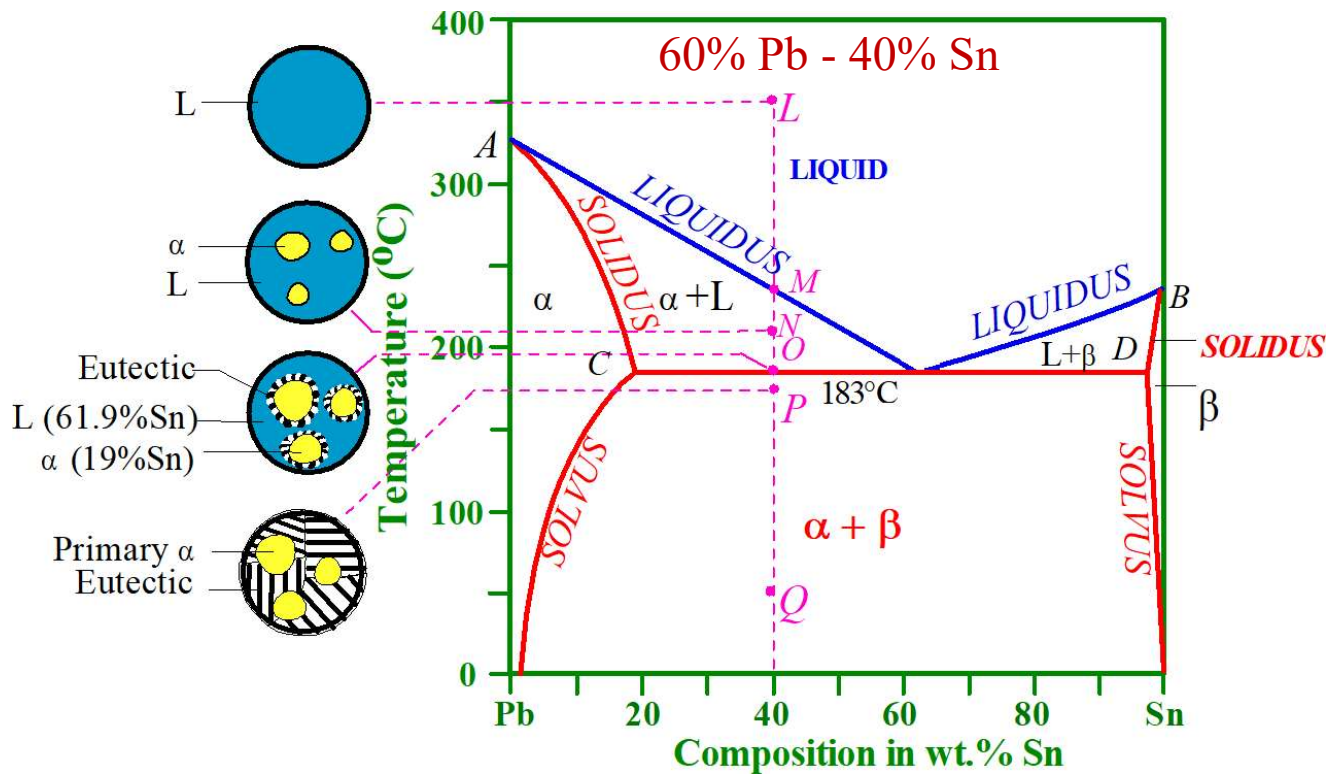


38.1% Pb-61.9% Sn Alloy



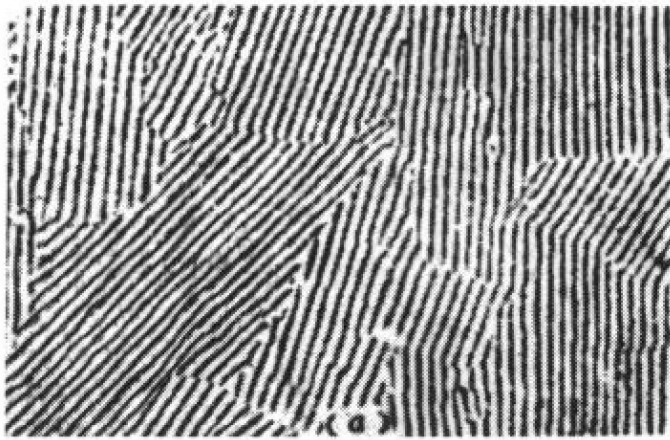
Pb-Sn alloy with Sn of 40% and 61.9%



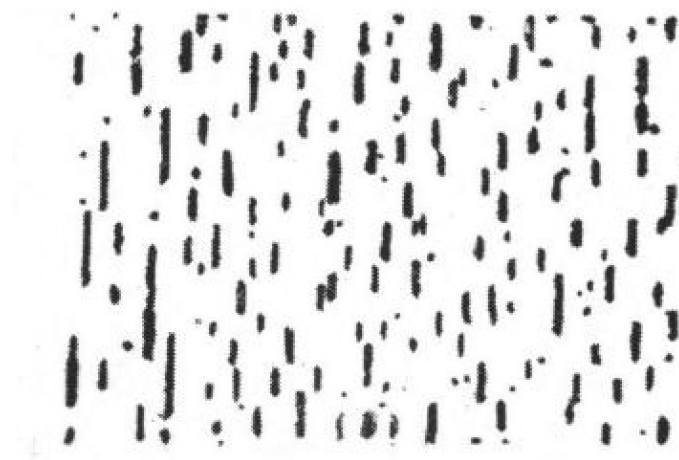


Temperature (°C)	Phases	Composition	Mass (g)	Microstructure and Comment
250	L	40% Sn	100	
235	L	40% Sn	100	The first solid (α -phase) nucleates in the liquid.
	α	15% Sn	0	
210	L	50% Sn	68.7	Mixture of liquid and α phases. More solid forms. Compositions change.
	α	18% Sn	31.3	
183.5	L	61.9% Sn	48.7	Liquid has the eutectic composition.
	α	19.2% Sn	51.3	
182.5	α	19.2% Sn	73.4	Eutectic (α and β phases) and primary α -phase.
	β	97.5% Sn	26.6	

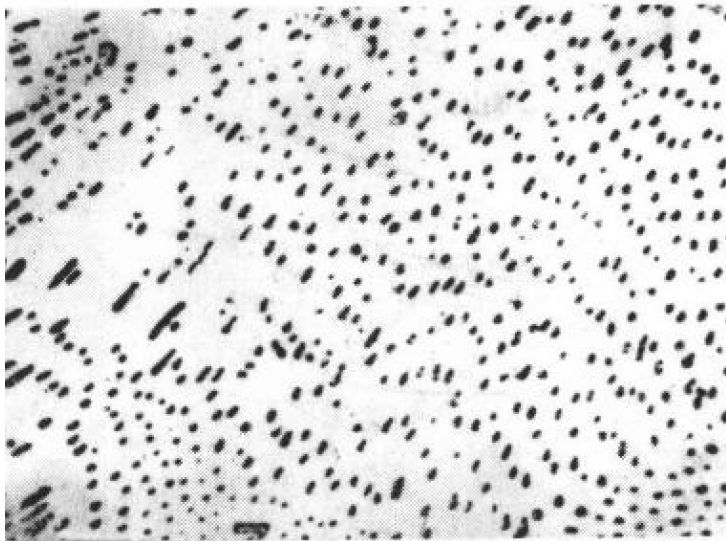




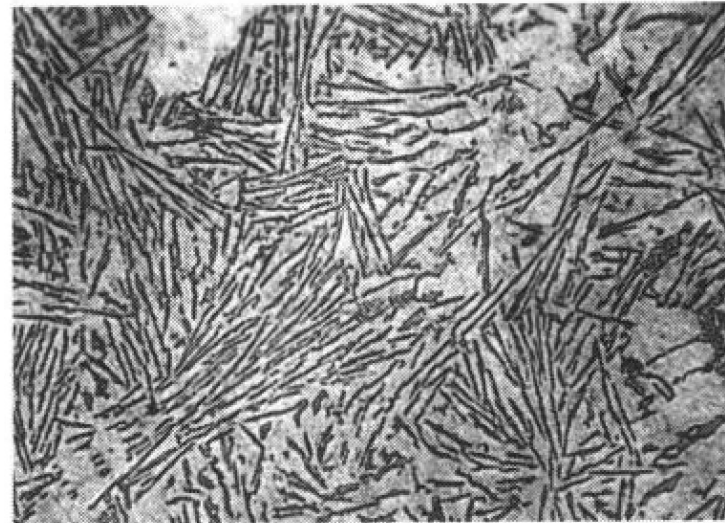
Sheet Structures of Pb-Sn Alloy
层片状 (Pb-Sn) 200X



Rod Shape
棒状

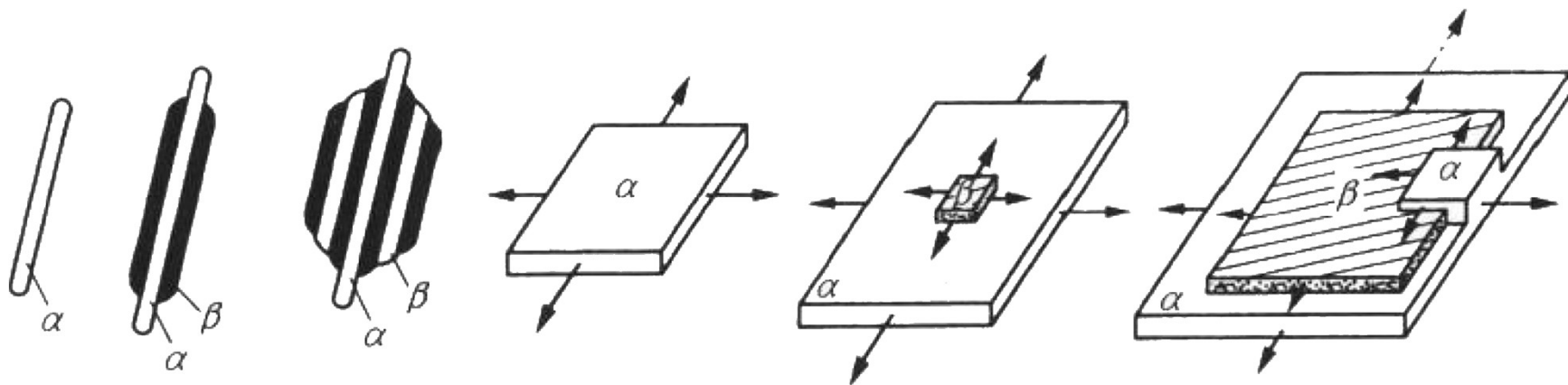


Spherical Structure of Cu-Cu₂O
球状 (Cu-Cu₂O)



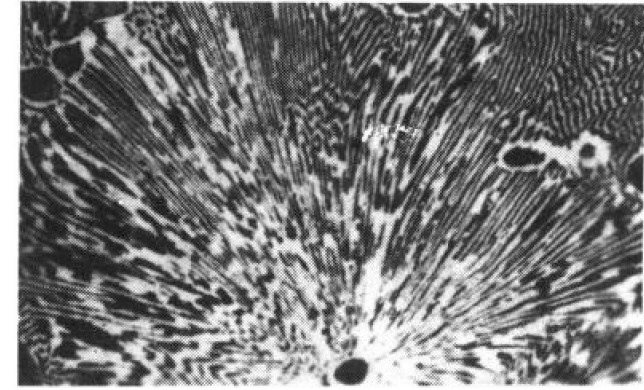
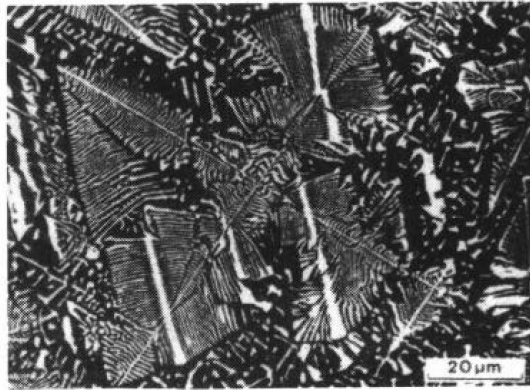
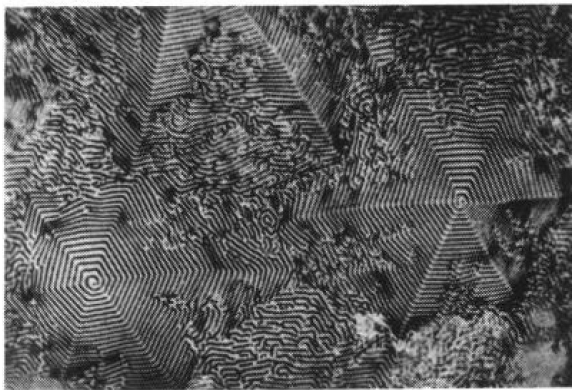
Needle Shape of Al-Si Alloy
针状 (Al-Si)





Layered alternating shape

Bridge structure



Spirals Structure, Spider Webs Structure
of Zn-MgZn Alloys

Radiative Structure of Cu-P Alloys

